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Demystifying Clone-Censor-Weighting to Studying Treatment Initiation Windows: An Example Using Publicly Available Synthetic Medicare Claims Data

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ABSTRACT

Background: Clone-censor-weighting (CCW) can compare treatment regimens that are initially indistinguishable (such as starting treatment within specific time windows) without using landmarks or creating immortal time. The causal contrasts estimated in these cases and the analyses themselves can become quite complex; however.

Objective: Provide a tutorial on CCW for comparing initiation windows and illustrate the causal contrasts underlying such comparisons.

Methods: We identified patients with myocardial infarctions without past aspirin or clopidogrel use in Medicare's synthetic public data files. We assigned “clones” to three regimens: (1) initiation within 30 days; (2) initiation within 90 days; or 3) initiation from 30 to 90 days. Clones were censored when deviating from their assigned regimen by failing to initiate treatment in time or by initiating treatment too early. We addressed informative censoring using inverse probability of censoring weights (IPCW), calculated weighted 180-day risks of re-hospitalization or death using Kaplan–Meier methods, and visualized the portion of the population exposed during the first 90 days to compare exposure distributions underlying regimens.

Results: We identified 1589 patients experiencing myocardial infarction with no past medication use. 15% initiated within 30 days and 26% initiated between 30 and 90 days. After IPCW, the 180-day outcome risk was 40.2% in the 30-day regimen, 35.7% in the 90-day regimen, and 35.2% in the 30-to-90 day regimen.

Conclusions: Though CCW can be complex to implement and the effects it estimates can vary substantially across study populations that initiate treatments at different times, it is a useful tool for contrasting initiation windows.

1 | Introduction

Researchers have mimicked randomized trials using nonexperimental data for decades. One persistent challenge is allocating time experienced by individuals while treatment regimens are indistinguishable from one another. Randomized trials can assign patients to different treatment durations or timings (e.g., 6 vs. 12 chemotherapy infusions) [1]. In observational data, however, such regimens are impossible to distinguish until long after treatment assignment (and may be

Summary

- Clone-censor-weighting (CCW) can compare the effects of different treatment initiation windows.
- The causal contrast estimated by CCW when studying initiation windows varies depending on when the population naturally initiates, making information on initiation timings essential context for estimates.
- We provide straightforward analytic code for a CCW analysis within a cohort constructed from publicly available synthetic Medicare claims data.
- We demonstrate how to construct censoring weights that avoid selection bias when exposure effects vary over time by only allowing the outcomes of recent initiators for estimating hypothetical outcomes of those censored at the end of windows.

completely impossible to distinguish, particularly in patients with early outcomes). Traditional observational analyses of these regimens (e.g., excluding immortal time from incidence rate calculations, performing landmark analyses, or using conditional estimates from time-varying exposure models) change estimated quantities vs. a hypothetical trial, require strong assumptions to estimate those quantities, misalign assessing eligibility, starting treatment, and beginning follow-up, or multiple of the above [2–5]. These issues also make these analyses difficult to incorporate into the increasingly popular target trial framework [6].

One solution involves cloning participants into regimens of interest, censoring clones once incompatible with assigned treatment regimens, and (until they are censored) weighting clones based on their inverse probability of remaining uncensored (IPCW) (hereafter referred to as CCW for clone-censor-weighting) [7, 8]. This corresponds to a hypothetical trial and aligns eligibility assessment, treatment start, and initiating follow-up. CCW can emulate trials of treatment regimens indistinguishable at baseline [9, 10] and compare outcomes when individuals receive treatment at different times (e.g., surveillance colonoscopy every 3 vs. 7 years) [11] or initiate treatment within windows (e.g., statin initiation within 6 months vs. no treatment [12] or starting hormone therapy at 3 months vs. at disease progression) [13].

Unfortunately, CCW analyses can be difficult to implement. Moreover, causal contrasts underlying CCW analysis comparing regimens like "start treatment within 30 days" to "start treatment from 30 to 90 days" are difficult to articulate because they can vary across study populations and information on the study population's initiation timings provides essential context for effect estimates [14]. Here, we walk through using CCW to estimate the effect of initiating aspirin or clopidogrel treatment within 30 days of first myocardial infarction (MI) vs. within 90 days vs. within 30-to-90 days in the Medicare Claims Synthetic Public Use Files (SynPUF) and visualize hypothetical interventions. The source cohort and commented SAS and R code are in a public GitHub repository (https://github.com/mawcpharmdphd/CCW_tutorial).

2 | Methods

2.1 | Data Source

SynPUF includes 3 years of synthetic data from the Centers for Medicare and Medicaid Services [15]. While variable associations are altered, SynPUF can illustrate applications of methods to claims data while allowing sharing of cohort files. Note that we excluded individuals from the base SynPUF when creating our cohort for the purposes of illustration.

2.2 | Causal Questions of Interest

We compared the effects of initiating aspirin or clopidogrel within 30 days of MI (hereafter, "0–30 day"), within 90 days ("0–90 day"), and from 30 to 90 days ("30–90 day") on 180-day risks of re-hospitalization or death in those naïve to aspirin and clopidogrel at the time of MI.

2.3 | Understanding the Hypothetical Interventions

While these regimens are easy to articulate ("start by day 30" vs. "start by day 90" vs. "start from day 30 to 90"), exposure patterns underlying them are deceptively complex. This is also true for other "representative interventions" or "natural interventions" relying on data structures due to grace periods (i.e., windows) [14, 16, 17]. There are 31 days people can start treatment by day 30, 91 days by day 90, and 61 days between day 30 and 90. The hypothetical intervention underlying "start by X" regimens is making the population follow its natural start time distribution until X, then making everyone who has not started treatment by X initiate at day X (in CCW analyses, this is done via those who initiate on day X "standing in" for everyone yet to initiate). Figure 1 illustrates how various initiation patterns within the study population can generate radically different hypothetical exposure patterns for the first 30 days of a "start by day 30" regimen. The hypothetical intervention underlying "start between X and Y" regimens is even more complex and involves preventing anyone from initiating prior to X, allowing people to naturally initiate between X and Y, and making those who have yet to initiate by Y to initiate treatment on Y.

Because differences in time exposed between regimens can vary across populations, comparisons of seemingly identical regimens may represent distinct causal questions across studies. Consider populations A and B. In A, 50% of the population initiates on day 29 and 50% initiates on day 31. In B, 50% initiates on day 0 and 50% initiates on day 90. In A, only 2 days of exposure in half the population separates "start by day 30" and "start between day 30 and 90." In B, on the other hand, half the population is exposed for the first 90 days in "start by day 30" while none are exposed for the first 90 days in "start between day 30 and 90." This makes information on initiation timings essential context for "start by X" regimens when exposure affects outcomes of interest.

2.4 | CCW Overview

CCW studies include five steps: identifying the population of interest, cloning and assigning clones to regimens, censoring clones after deviation from assigned regimens, weighting clones to address bias arising from informative censoring, and analyzing the weighted cohort (Table 1). Figure 2 describes the first four steps in the context of our case study.

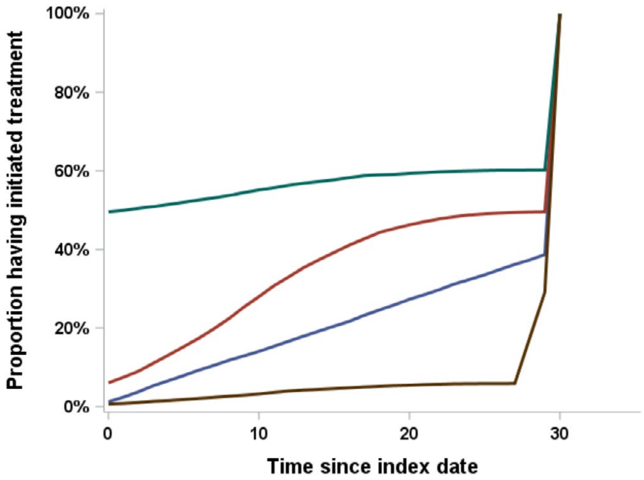

FIGURE 1 | A plot of the proportion of the population that has initiated a treatment by each date from day 0 to day 30 in a CCW analysis of a "start by day 30" treatment regimen in various populations. Each line corresponds to a specific type of treatment distribution (brown = late start, green = early start, red = start times vary based on a normal distribution, blue = start times follow a uniform distribution). In every population, there is a sudden spike on day 30 when those who have yet to initiate start treatment under the hypothetical intervention, but prior to that the proportion of the population exposed to the treatment varies substantially.

2.5 | Step 1—Identifying a Study Cohort

First, specify potential index dates, eligibility criteria, and outcomes. In our example, the potential index date is hospital discharge with an ICD-9 diagnosis code for acute MI in the primary position. After identifying potential index dates, apply eligibility criteria: first, no prior codes for acute MI; second, no use of aspirin or clopidogrel in the year prior to the MI; and third, prescription coverage during the year of the MI. We should also identify the date of the next hospitalization or death, whichever is earliest, as well as the first aspirin or clopidogrel prescription fill date (if it exists). This generates a time-to-event cohort.

2.6 | Step 2—Cloning

Next, create copies ("clones") to assign to regimens of interest. In the example, we make three clones and assign them to "0–30 day," "0–90 day," and "30–90 day" regimens. We must decide how to handle patients incompatible with regimens at baseline (e.g., patients with a prescription for aspirin on the day of discharge are incompatible with the "30–90 day" regimen). When using CCW to estimate per-protocol effects of continuing a treatment A for a specific duration in trials that randomized patients to initiate treatment A or placebo, there is no need to clone "incompatible" individuals [18]. In observational studies, however, failing to clone them can generate imbalances between regimens (i.e., confounding) [19]. We can avoid baseline imbalances by cloning everyone into all regimens and creating "time zero" censoring weights, by not cloning into incompatible regimens and creating separate treatment weights (with weights deciding the target population), or excluding those incompatible with a treatment regimen completely. To maintain consistent target populations across comparisons while maximizing sample size, we cloned everyone into all regimens and created a special "time 0 censoring" flag. An alternative to cloning patients

TABLE 1 | Essential steps when using cloning, censoring, and weighting to study treatment initiation windows.

Step	Description	Key considerations
Identification	Identify index dates where patients meet eligibility criteria to establish time 0 for each study participant and an overall target population of interest	Make sure to address whether individuals can meet eligibility criteria at multiple dates
Cloning	Create copies (clones) of each eligible participant and assign each clone to one of the initiation windows under study beginning at that patient's index date	Carefully consider the length of windows Be cautious with "day 30 or later" windows
Censoring	Censor each clone once they are incompatible with their assigned treatment regimen, whether due to starting treatment outside the treatment window or due to failing to start by the end of the window	Important to specify and code reasons for censoring, especially when there are multiple censoring mechanisms
Weighting	Create person-period data sets and assign time-varying inverse probability weights based on the probability of remaining uncensored during each person-period, conditional on measured variables	Fit separate weights for each censoring mechanism Weigh how many assumptions you want to make about model form
Analysis	Analyze the weighted person-period data to obtain the effect estimate of interest	Make sure variance is treated properly by statistical software Recommend some visualizations of treatment timing distributions

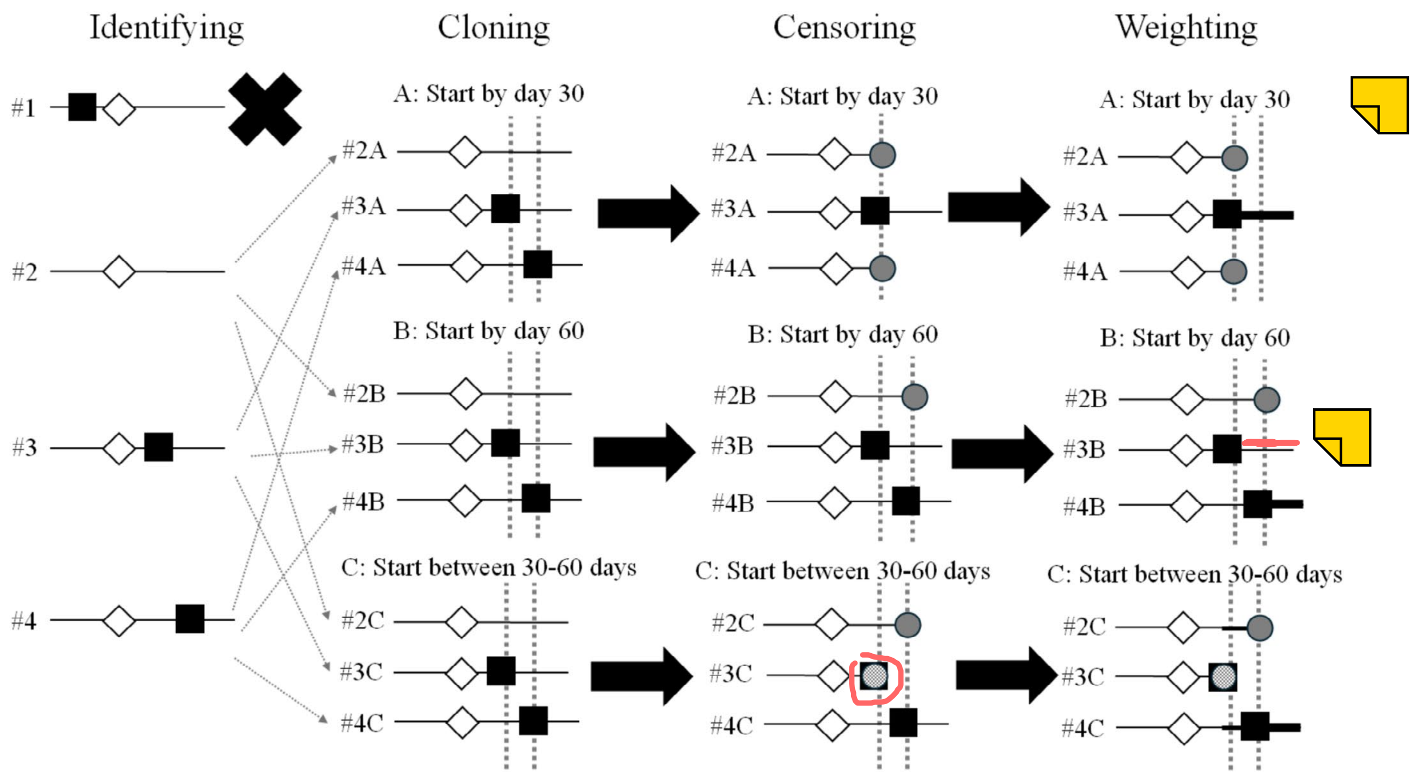

FIGURE 2 | A figure describing the first four steps of a clone-censor-weighting analysis in the context of comparing effects of different treatment initiation windows. White diamonds represent potential index dates and black squares represent prescription records. After identifying eligible individuals (in this case, by excluding the individual with treatment prior to their potential index dates), duplicates, or clones, are created within each treatment regimen of interest (i.e., start by day 30, start by day 60, and start between 30 and 60 days). Those clones are then censored when they deviate from their assigned regimen either by failing to initiate by the end of the window (represented by solid gray circles) or, for the start between 30- and 60-day regimen, by initiating before day 30 (represented by the checkered gray circle). Gray dashed lines indicate 30 and 60 days post index. Weighting (represented by thicker lines) is then used to upweight those who were not censored to "stand-in" for those who were. Note that in regimen B person 4, who initiated close to day 60, is upweighted to stand in for the person censored for not initiating by day 60, while person 3 receives a weight of 1 on day 60 and later.

into all regimens is randomizing patients to one regimen. This may be faster when data sets are large and precision is not a concern. After randomization, analysis proceeds identically. Due to low treatment rates, we are using CCW to maximize efficiency.

2.7 | Step 3—Censoring

Next, censor clones. Censoring clones reduces follow-up to when they met censoring criteria, sets their outcome to 0, and assigns a censoring flag. Censoring criteria can vary across regimens but, for initiation windows, will generally occur due to starting treatment too early or failing to start treatment by the end of the window. In the "0–30 day" regimen, for example, observations can only be censored on day 30 for not initiating treatment. In the "30–90 day" regimen, patients can be censored for initiating between day 0 and 30 (getting a censoring flag for starting early) or for failing to start on day 90 (getting a censoring flag for not starting).

Censoring flags identify censoring reasons and are specific to each type of censoring; in the "30–90 day" regimen, for example, there should be separate flags for censoring due to starting before day 30 and failing to start by day 90. Note that if there are other censoring criteria researchers should consider (e.g., discontinuing or switching treatment following initiation, switching insurance plans, other loss to follow-up), additional flags are necessary. This results in three time-to-event data sets with one observation per individual with censored follow-up times, outcomes, and censoring flags.

2.8 | Step 4—Weighting

If there are factors associated with both censoring and the outcome, censoring will generate selection bias [20–22]. If older individuals are more likely to initiate prior to day 30 than younger individuals, for example, a disproportionate number of older individuals remain post day 30 after censoring. To address this, when censoring occurs we weight remaining person-time of the uncensored participants by the inverse of their probability of remaining uncensored (i.e., time-varying inverse probability of censoring weights, IPCW). This involves creating longitudinal data sets and creating weights within each treatment regimen.

2.9 | Weighting Regimens With Intervals Including Time 0

Assuming no other informative censoring, creating IPCW for regimens with windows beginning at time 0 (e.g., "0–30 day" and "0–90 day") requires up to two observations per patient, one during the window (e.g., 0–30 days for "0–30 day") and one starting when the window closes (e.g., day 30+ for "0–30 day"). Patients with follow-up ending during the window (due to administrative censoring or outcomes) receive one observation. Patients followed longer than the window that initiate during the window receive two observations, one covering the window and one starting when the window closes and lasting until the end of follow-up. Patients followed longer than the window that do not initiate within the window generate two observations: one for the window and one when the window closes with 0 days of follow-up (to calculate weights). Importantly, time-varying covariates (e.g., age) are updated for each observation.

We then need to estimate the probability of remaining uncensored for failing to start treatment by the start of the second interval conditional on observed time-updated values of covariates L (here, age, sex, and renal disease), as well as past exposure history X_{hist} (i.e., at what time if any they started treatment during the interval). Uncensored second observations receive weights equal to 1 divided by this probability, or:

$$IPCW_i = 1 / \left(\Pr\left(Censor_{nostart} = 0 | L, X_{hist}\right)\right)$$

We can estimate cumulative probabilities of remaining uncensored in those unexposed before the final day of the interval using multivariable logistic regression (the method we will demonstrate) or a Cox model treating censoring as an "event." Note that all individuals who initiate treatment prior to the final day have a 100% probability of remaining uncensored on day 30 (e.g., everyone who received aspirin before day 29 will be uncensored on day 30), meaning they should technically receive weights of 1.

While it may seem intuitive to give those who initiated early in the window IPCW weights besides 1, that approach allows day 0 initiators to stand in for people censored at day 30 (see Supporting Information: Appendix 1). If exposure effects are time-varying or exposure has an effect on the covariates used in day 30 IPCW, this biases estimates. Unfortunately, including only those who initiate on the final day of the interval may not be feasible (in our case, there are 5 initiators on day 30 and 4 on day 90). To address this without completely ignoring time-varying effects, we fit models in "recent initiators" who initiated near the end of the interval (i.e., after day 23 or 83). As this approach introduces selection bias if exposure effects are time-varying, definitions of "recently starting" should be varied in sensitivity analyses.

2.10 | Weighting Regimens With Intervals Starting After Time 0

Weighting for the "30–90 day" regimen requires separate IPCW to address censoring due to both starting early and failing to start by day 90. First, address day 0 initiators by creating an IPCW model at time 0, ensuring the baseline population after accounting for those that are immediately censored reflects the target population of study-eligible patients. Next, handle censoring between day 1 and 30 for starting treatment. The simplest approach is breaking days 0–29 into one-day intervals, predicting censoring each day with a pooled multivariable logistic regression model flexibly adjusting for the time at the start of the interval as well as time-updated covariate values [23]. Pooling intervals avoids sparse data issues but requires parametric assumptions about how censoring varies over time. Longer intervals avoid computational issues in large cohorts but risk potential model misspecification.

2.11 | Final Considerations

The above interval-specific IPCW are ultimately combined into cumulative IPCW (corresponding to the cumulative survival probabilities if a Cox model is used to create IPCW) by multiplication to account for all censoring of a specific type up to the start of the interval. It is important to note that if other informative censoring occurs, additional observation periods (and additional censoring-reason-specific IPCW), similar to those addressing "start early" censoring in the "day 30–90" regimen, are needed, and these other types of informative censoring will need to be addressed with separate censoring-reason-specific IPCW to be incorporated into the cumulative IPCW (Weuve et al. Epidemiology, 2012).

2.12 | Step 5—Analysis of the Weighted Data

Analyses can be performed using any approaches suitable for weighted right-censored data. We used weighted Kaplan–Meier methods to estimate 180-day risks under each regimen and calculated risk differences. Obtaining confidence intervals requires bootstrapping [24].

Analysis also includes visualizing exposure patterns underlying regimens and evaluating IPCW performance. We estimated cumulative proportions of the population exposed in each regimen, calculated standardized mean differences (SMDs) in time-updated covariates at day 30 (in '0–30 day") or day 90 (in "0–90 day" and "30–90 day") between recent initiators and the population they are "standing in for" (i.e., the population prior to censoring except for those who initiated prior to the start of the "recent initiation" window) and compared the number of the latter group to the number of recent initiators after weighting to identify positivity violations [25].

3 | Results

We identified 32,591 MIs in SynPUF and 1589 patients met eligibility criteria. Of those, 234 (15%) initiated aspirin or clopidogrel within 30 days and 411 (26%) additional patients initiated treatment within 90 days. Five initiated treatment on day 0. There were 593 outcomes within 180 days.

The "0–30 day" regimen censored 1355 patients for failing to start by day 30, while the "0–90 day" and "30–90 day" regimens censored 944 patients for failing to start by day 90. The "30–90 day" regimen censored 229 patients for starting treatment early.

After IPCW, 180-day outcome risk was 40.2% in the "0–30 day" regimen, 35.7% in the "0–90 day" regimen, and 35.2% in the

"30–90 day" regimen. This resulted in risk differences of −4.6% and −5.0% comparing "0–90 day" and "30–90 day" regimens to the "0–30 day" regimen, respectively.

Figure 3 provides cumulative probabilities of initiation between day 0 and 90 within regimens after IPCW. Initiation rates were consistent throughout. Exposure patterns during the first 30 days were identical for the "0–30 day" and "0–90 day" regimens and initiation rates (i.e., the slope of the line) were higher for the "30–90 day" regimen than the "0–90 day" regimen from day 30 to 90.

Table 2 shows SMDs between the pre-censoring study participants minus those who initiated prior to the start of the recent initiation window and recent initiators before and after applying IPCW. Absolute SMDs are less than 0.100 after IPCW and covariates are better balanced after applying weights. The total number of initiators after weighting ("0–30 day" recent initiator weighted N: 1254.2; "0–90 day" weighted N: 766.3; "30–90 day" weighted N: 896.4) and the pre-weight cohort the weights were constructed in ("0–30 day" total N: 1254, "0–90 day" censored N: 765, "30–90 day" censored N: 893.9) were almost identical, suggesting minimal positivity violations.

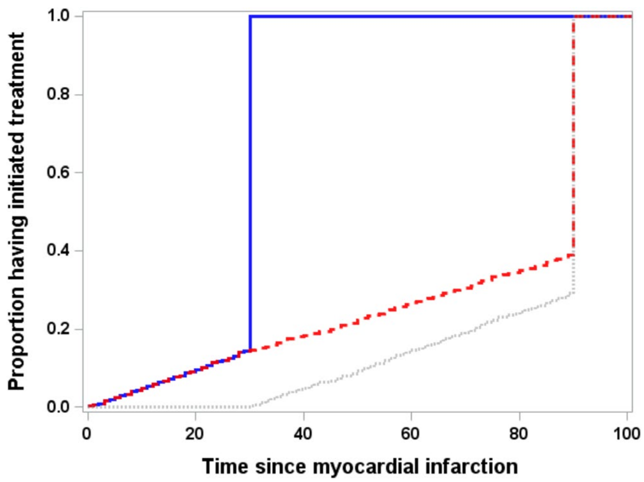

FIGURE 3 | Proportion of the clones assigned to the "0–30 day" (solid blue line), "0–90 day" (dashed red line), and "30–90 day" (dotted gray line) regimens.

4 | Discussion

We guided readers through a CCW analysis comparing initiation windows for aspirin and clopidogrel in patients after MI in synthetic Medicare claims data. We discussed the complex hypothetical interventions under study, whether to clone those incompatible with their regimens at time 0, and reasons to include only recent initiators when creating IPCW at the end of the treatment window. Finally, we included code visualizing differences in exposure patterns between regimens to provide context to the underlying exposure patterns as well as an example of how to examine censoring weight performance.

Of course, CCW is not a silver bullet. Interventions underlying "start by day X" and "start between day X and Y" regimens vary depending on initiation patterns within the study population. As a result, interpreting effect estimates or comparing findings across different studies studying the same regimens requires information on initiating timings (similar to studies of other "natural" interventions) [14]. Additionally, if predictors of censoring associated with the outcome are unmeasured or omitted from IPCW, estimates will be biased. Researchers must also be concerned about exposure, covariate, and outcome measurement error [26], differential surveillance between individuals [27],

TABLE 2 | Covariate distributions of, and SMDs comparing, the pre-censoring population minus those who initiated prior to the recent initiation window and the recent initiators before and after applying IPCW.

Regimen and covariate	All individuals at the censoring timing	Recent initiators	SMD before IPCW for failing to initiate	SMD after IPCW for failing to initiate
N for "0–30 day" regimen	1198	56		
Age (mean, S.D.)	73.1 (13.2)	72.2 (13.7)	−0.074	0.002
Female sex *N* (%)	497 (41%)	19 (34%)	−0.119	−0.001
Renal failure *N* (%)	368 (31%)	16 (29%)	−0.017	−0.027
N for "0–90 day" regimen	718	47		
Age (mean, S.D.)	73.3 (13.1)	72.7 (15.0)	−0.048	−0.004
Female sex *N* (%)	299 (42%)	20 (43%)	0.071	−0.010
Renal failure *N* (%)	206 (29%)	11 (23%)	−0.081	0.006
N for "30–90 day" regimen[a]	893.9	54.9		
Age (mean, S.D.)	73.2 (13.1)	72.5 (15.0)	−0.053	−0.016
Female sex *N* (%)	346.3 (39%)	23.0 (42%)	0.066	−0.014
Renal failure *N* (%)	245.1 (27%)	13.2 (24%)	−0.097	0.012

Abbreviations: IPCW = inverse probability of remaining uncensored weights, S.D. = standard deviation, SMD = standardized mean difference.
[a]Values for the "30-90 day" regimen were calculated using the cumulative IPCW for initiating on day 0 to 29.

model misspecification [28], missing data [27], and other common sources of bias [29].

This article focused on providing readers with tools for performing, understanding, and replicating straightforward CCW studies contrasting initiation windows. Requiring individuals to remain on treatment after initiation or comparing effects of initiating different treatment options introduces additional analytic complexity [30]. We also contrasted effects of different windows rather than comparing windows to never initiating, though code to analyze a "never initiate" regimen, which expands the code covering the first 30 days of the "30–90 day" regimen to the full 180-day period, is also available. Models for the probability of remaining uncensored were also fit within the clones assigned to each treatment strategy, rather than within the base cohort; under additional parametric assumptions about treatment initiation probabilities, fitting models within the base cohort can increase precision by drawing on information about the characteristics of people who initiated outside a specific regimen's treatment window [17].

5 | Conclusion

While CCW studies can be challenging to conduct and their findings must be considered in the context of initiation patterns within cloned populations, they represent a powerful tool for studying the effects of initiating treatment within specific time windows.

5.1 | Plain Language Summary

Researchers have recently started using a new method, called "clone-censor-weighting" (CCW), to compare the effect of starting drug or medical treatments during specific time windows (e.g., between 0 and 30 days or between 30 and 90 days after a heart attack). Using CCW to make these comparisons can be very complicated, however, both in terms of actual implementation and in terms of the actual questions researchers are answering. We walk readers through a fairly simple application of the CCW method to compare the effect of starting aspirin or clopidogrel within 30 days, within 90 days, or between 30 and 90 days of a heart attack within fake data originally created by the Centers for Medicare and Medicaid Services. We also provide statistical code for readers to duplicate our analyses. In addition to removing bias from measured variables that impacted whether people were removed from our study, we also created figures to represent when people started treatment within each of the hypothetical treatment regimens being studied. While using CCW can be challenging and it answers a deceptively complicated research question, it is very useful for comparing different windows in which people might start treatment.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Medicare Claims Synthetic Public Use files are freely available online. SAS code is included in the appendix of this article, along with the base analytic cohort, are on GitHub at https://github.com/mawcpharmdphd/CCW_tutorial.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.